

SE Lab 4

Before Video

https://drive.google.com/file/d/1fUrF2GO_i7G3lpc0XZWM0jNpH4Y5k9Xr/view?usp=sharing

After Video

https://drive.google.com/file/d/13gCY5nViVAqINijwiOMxB81-mwg_MEnH/view?usp=sharing

ChatGPT Link: <https://chatgpt.com/share/6ac79529-1c38-83ee-9c27-f8724a9a43c9>

Forked Repository: https://github.com/riddhi2106/04_target_shooting

Code Snippets

Task 1

```
target-shooting > game > hit_detection.py > check_hit
1  """
2  hit_detection: figures out whether a click landed on a target.
3  """
4
5
6  def check_hit(targets, click_pos):
7      """
8      Returns the target that was clicked, or None if the click missed
9      every target.
10     """
11     click_x, click_y = click_pos
12
13     for target in targets:
14         dx = click_x - target.x
15         dy = click_y - target.y
16
17         if dx * dx + dy * dy <= target.radius * target.radius:
18             return target
19
20     return None
```

Task 2

```
target-shooting > game > target.py > ...
1  """
2  Target: a circular target the player clicks on. (x, y) is the CENTER
3  of the circle - this matters for how it's drawn vs. how it's hit-tested.
4  """
5
6  import pygame
7
8
9  class Target:
10     def __init__(self, x, y, radius=28, color=(230, 90, 70), speed=2):
11         self.x = x
12         self.y = y
13         self.radius = radius
14         self.color = color
15         self.speed_x = speed
16         self.speed_y = speed
17
18     def get_bounding_rect(self):
19         """A square bounding box around the circle."""
20         return pygame.Rect(self.x, self.y, self.radius * 2, self.radius * 2)
21
```

```
def _random_target(self, index=0):
    x = random.randint(TARGET_RADIUS + 10, WIDTH - TARGET_RADIUS - 10)
    y = random.randint(TARGET_RADIUS + 10, HEIGHT - TARGET_RADIUS - 10)

    speed = 2 if index % 2 == 0 else 4

    return Target(x, y, radius=TARGET_RADIUS, speed=speed)

def handle_click(self, pos):
    if self.round_over:
        return

    target = check_hit(self.targets, pos)

    if target is not None:
        self.combo += 1
        self.score += POINTS_PER_COMBO * self.combo

        self.targets.remove(target)
        self.targets.append(self._random_target(len(self.targets)))
    else:
        self.combo = 0
```

```
def update(self):
    if self.round_over:
        return

    elapsed = (pygame.time.get_ticks() - self.start_time) / 1000
    self.time_remaining = max(0, ROUND_DURATION - elapsed)

    if self.time_remaining <= 0:
        self.time_remaining = 0
        self.round_over = True
        return

    for target in self.targets:
        target.x += target.speed_x
        target.y += target.speed_y

        if target.x - target.radius <= 0:
            target.x = target.radius
            target.speed_x *= -1
        elif target.x + target.radius >= WIDTH:
            target.x = WIDTH - target.radius
            target.speed_x *= -1

        if target.y - target.radius <= 0:
            target.y = target.radius
            target.speed_y *= -1
        elif target.y + target.radius >= HEIGHT:
            target.y = HEIGHT - target.radius
            target.speed_y *= -1
```

Task 3

```
def draw(self, surface, font):
    from game import renderer

    renderer.draw_scene(surface, self.targets)

    renderer.draw_text(
        surface,
        font,
        f"Score: {self.score} Combo: x{self.combo}",
        (10, 10)
    )
```

Task 4

```
renderer.draw_text(
    surface,
    font,
    f"Time: {self.time_remaining:.1f}s",
    (10, 40)
)

if self.round_over:
    renderer.draw_banner(
        surface,
        font,
        f"TIME'S UP! Final Score: {self.score} | Press R to restart"
    )
```

```
def main():
    pygame.init()
    screen = pygame.display.set_mode(WINDOW_SIZE)
    pygame.display.set_caption("Target Shooting")
    clock = pygame.time.Clock()
    font = pygame.font.SysFont("consolas", 22)

    engine = GameEngine()
    running = True
    while running:
        for event in pygame.event.get():
            if event.type == pygame.QUIT:
                running = False
            elif event.type == pygame.MOUSEBUTTONDOWN:
                engine.handle_click(event.pos)

            elif event.type == pygame.KEYDOWN:
                if event.key == pygame.K_r and engine.round_over:
                    engine.start_round()

        engine.update()
        engine.draw(screen, font)

        pygame.display.flip()
        clock.tick(60)

    pygame.quit()
```